Problem Set 3

Applied Stats II

Due: March 25, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in `.pdf` form.
- This problem set is due before 23:59 on Wednesday March 25, 2026. No late assignments will be accepted.

Question 1

We are interested in how governments' management of public resources impacts economic prosperity. Our data come from Alvarez, Cheibub, Limongi, and Przeworski (1996) and is labelled `gdpChange.csv` on GitHub. The dataset covers 135 countries observed between 1950 or the year of independence or the first year forwhich data on economic growth are available ("entry year"), and 1990 or the last year for which data on economic growth are available ("exit year"). The unit of analysis is a particular country during a particular year, for a total $>$ 3,500 observations.

- Response variable:
  - `GDPWdiff`: Difference in GDP between year t and $t-1$. Possible categories include: "positive", "negative", or "no change"
- Explanatory variables:
  - `REG`: 1=Democracy; 0=Non-Democracy
  - `OIL`: 1=if the average ratio of fuel exports to total exports in 1984-86 exceeded 50%; 0= otherwise

Please answer the following questions:

1. Construct and interpret an unordered multinomial logit with `GDPWdiff` as the output and "no change" as the reference category, including the estimated cutoff points and coefficients.

```
gdp_data$GDPWdiff_cat <- relevel(gdp_data$GDPWdiff_cat, ref = "no change")

unorder_gdp <- multinom(
  GDPWdiff_cat ~ REG + OIL,
  data = gdp_data,
  MaxNWts = 10000
)
summary(unorder_gdp)
```

Table 1: Unordered Multinominal Logit

	Dependent variable:	
	negative	positive
	(1)	(2)
REG	1.379*	1.769**
	(0.769)	(0.767)
OIL	4.784	4.576
	(6.885)	(6.885)
Constant	3.805***	4.534***
	(0.271)	(0.269)
Akaike Inf. Crit.	4,690.770	4,690.770
Note:	*p<0.1; **p<0.05; ***p<0.01	

- In a given country, there is an increase in the baseline odds (no change in GDP) that the difference in GDP will be negative by 1.379 times when a country is a democracy.
- In a given country, there is an increase in the baseline odds (no change in GDP) that the difference in GDP will be positive by 1.769 times when a country is a democracy.
- In a given country, there is an increase in the baseline odds (no change in GDP) that the difference in GDP will be negative by 4.784 times given the average ratio of fuel exports in 1984-86 exceeded 50 percent.

- In a given country, there is an increase in the baseline odds (no change in GDP) that the difference in GDP will be postive by 4.576 times given the average ratio of fuel exports in 1984-86 exceeded 50 percent.

2. Construct and interpret an ordered multinomial logit with `GDPWdiff` as the outcome variable, including the estimated cutoff points and coefficients.

```
order_gdp <- polr(GDPWdiff_cat ~ REG + OIL, data = gdp_data, Hess = T)
gdp_conf <- exp(cbind(OR = coef(order_gdp), confint(order_gdp)))

stargazer(order_gdp, type = "latex", title = "Ordered Multinominal")
stargazer(gdp_conf, type = "latex")
```

Table 2: Ordered Multinominal Logit

	Dependent variable:
	GDPWdiff_cat
REG	0.410***
	(0.075)
OIL	−0.179
	(0.115)
Observations	3,721
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 3: Odds Ratio and CI

	OR	2.5 %	97.5 %
REG	1.507	1.301	1.747
OIL	0.836	0.668	1.051

- For a country that is a democracy, the log odds of being in a higher GDP change category are 1.507 times higher compared to a country that isn't a democracy, holding all other variables constant.
- IF the coefficent was statisically significant: For a country that had the average ratio of fuel exports to total exports in 1984-86 that exceeded 50 percent, the log odd of being in a higher GDP change category are 0.836 times higher compared to a country that did not, holding all other variables constant.

Question 2

Consider the data set `MexicoMuniData.csv`, which includes municipal-level information from Mexico. The outcome of interest is the number of times the winning PAN presidential candidate in 2006 (`PAN.visits.06`) visited a district leading up to the 2009 federal elections, which is a count. Our main predictor of interest is whether the district was highly contested, or whether it was not (the PAN or their opponents have electoral security) in the previous federal elections during 2000 (`competitive.district`), which is binary (1=close/swing district, 0="safe seat"). We also include `marginality.06` (a measure of poverty) and `PAN.governor.06` (a dummy for whether the state has a PAN-affiliated governor) as additional control variables.

(a) Run a Poisson regression because the outcome is a count variable. Is there evidence that PAN presidential candidates visit swing districts more? Provide a test statistic and p-value.

- Null hypothesis: There is no difference in PAN visits in whether the district is a close/swing or "safe seat".
- Alternative hypothesis: There are more PAN visits to a swing district

```r
PAN_pois <- glm(PAN.visits.06 ~ competitive.district + marginality.06 +
    PAN.governor.06,
                data = mexico_elections,
                family = poisson)
summary(PAN_pois)
summary(PAN_pois)$coefficients["competitive.district", ]
#   Estimate   Std. Error      z value    Pr(>|z|)
#  -0.08135181  0.17068819 -0.47661062  0.63363942
```

Table 4:

	Dependent variable:
	PAN.visits.06
competitive.district	−0.081
	(0.171)
marginality.06	−2.080***
	(0.117)
PAN.governor.06	−0.312*
	(0.167)
Constant	−3.810***
	(0.222)
Observations	2,407
Log Likelihood	−645.606
Akaike Inf. Crit.	1,299.213
Note:	*p<0.1; **p<0.05; ***p<0.01

The p value of 0.63 for the coefficent competitive.district is greater than 0.05, therefore, we fail to reject the null hypothesis that there is no difference between PAN visits in whether the district is a close/swing or "safe seat". There is insufficent evidence to support the alternative hypothesis that there are more PAN visits in swing districts.

(b) Interpret the `marginality.06` and `PAN.governor.06` coefficients.

- An increase of marginality by 1 unit decreases the number of visits by PAN presidential canidate by a multiplicative factor of $\exp -2.080 \approx 0.125$.
- When a district has a PAN-affiliated governor, it decreases the number of visits by PAN presidential canidate by a multiplicative factor of $\exp -0.312 \approx 0.73$.

(c) Provide the estimated mean number of visits from the winning PAN presidential candidate for a hypothetical district that was competitive (`competitive.district`=1), had an average poverty level (`marginality.06` = 0), and a PAN governor (`PAN.governor.06`=1).

$$log(\lambda) = -3.810 - 0.081(competitive.district) - 2.080(marginality.06) - 0.312(PAN.governor.06) \quad (1)$$

$$log(\lambda) = -3.810 - 0.081(1) - 2.080(0) - 0.312(1) \quad (2)$$

$$log(\lambda) = -4.203 \quad (3)$$

The mean number of visits from the winning PAN presidential canidate for a district that was competitive, had an average poverty level and a PAN governor is -4.203. Since the problem is dealing with visits, it is impossible to visit a negative amount of times to a district so it can be said that the number of visits is 0.